



SBL **ENERGY** LIMITED



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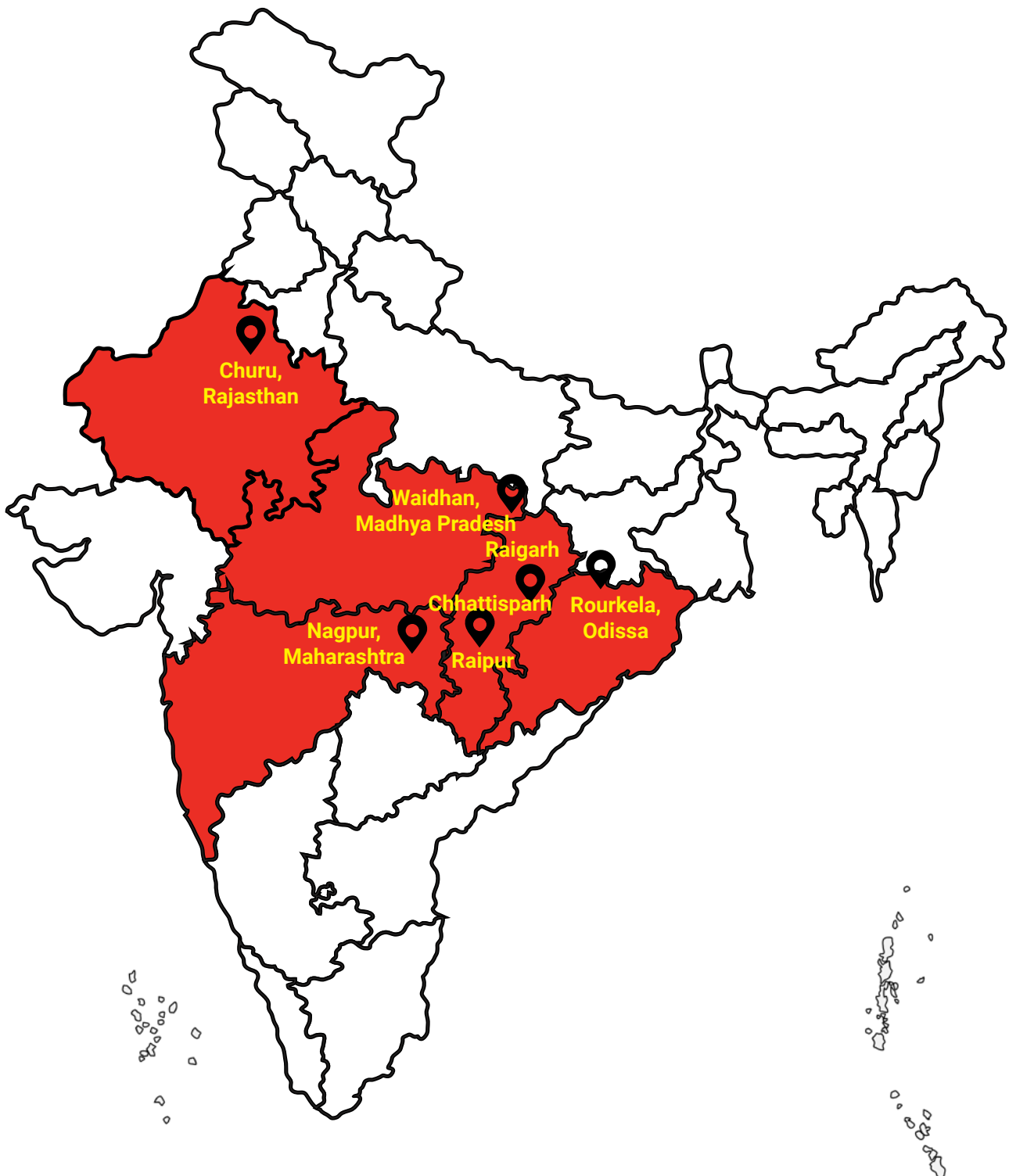
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STRATEGIC MANUFACTURING LOCATIONS



COMPANY PROFILE

SBL Group

Leading India's explosives industry with contemporary state of art manufacturing facility which ensures sustainable quality and concurring innovations - serving mining, energy, infrastructure and defense sector.

Who we are

SBL Group is trusted technical service provider and manufacturer of commercial explosives and blasting accessories, serving the mining, energy and infrastructure sectors for 3 decades. Group has also extended its footprints in defense sector.

Scale & people

We operate state-of-the-art production facilities across multiple strategic locations, supported by 2000+ professionals and a dedicated mining technical support team of 150 professionals.

Market presence

- Partner to India's largest mining corporations with an expanding international footprint across 22+ countries
- Market share: 12% in India and sizable market share globally
- **What we deliver** - End-to-end solutions spanning products and on-ground blasting expertise-built for safety, reliability, and performance.

VISION

To be the global benchmark for safety and reliability in energetic materials, recognized for redefining excellence through innovation and a steadfast commitment to a safer, more secure world.

MISSION

Our mission is to set the global standard in the production of high-performance explosives and defense products. Through a culture of continuous R&D and operational excellence, we deliver superior quality and value-driven solutions, ensuring our clients receive world-class expertise and timely delivery without compromise.

VALUES

At SBL Group, we are guided by five core principles:
Absolute Integrity | Safety First | Innovative Thinking
Operational Excellence | Social Responsibility

MANUFACTURING EXCELLENCE



World-class production capabilities (annual capacity)

Explosives

300,000+

Metric tons

Detonators

200 million+

Units

Detonating Fuse

150 million+

Meters

Cast boosters

5,000+

Metric tons

TNT

15,000+

Metric tons

RDX

1,000+

Metric tons

HMX

1,000+

Metric tons



PRODUCT OFFERINGS

Every product undergoes stringent testing against international standards to ensure absolute safety and quality.

COMMERCIAL EXPLOSIVES

- Cartridge Emulsion Explosives
- Bulk Emulsion Explosives
- Seismic Explosives

INITIATING SYSTEMS / ACCESSORIES

- Detonating Fuse / Cord
- Cast Booster
- Ordinary Detonators
- Elemented Caps
- Non-electric Detonators
- Electronic Detonators
- Shock Tubes

DEFENSE PRODUCTS

- TNT
- HMX
- RDX
- RDX and HMX Compound
- Pyro Devices
- Ammunition

BLASTING CONSULTING & TECHNICAL SERVICE

- Explosive Selection
- Blast Design
- Blast Management and Optimisation
- Explosive Regulatory Compliance

OTHER CHEMICALS

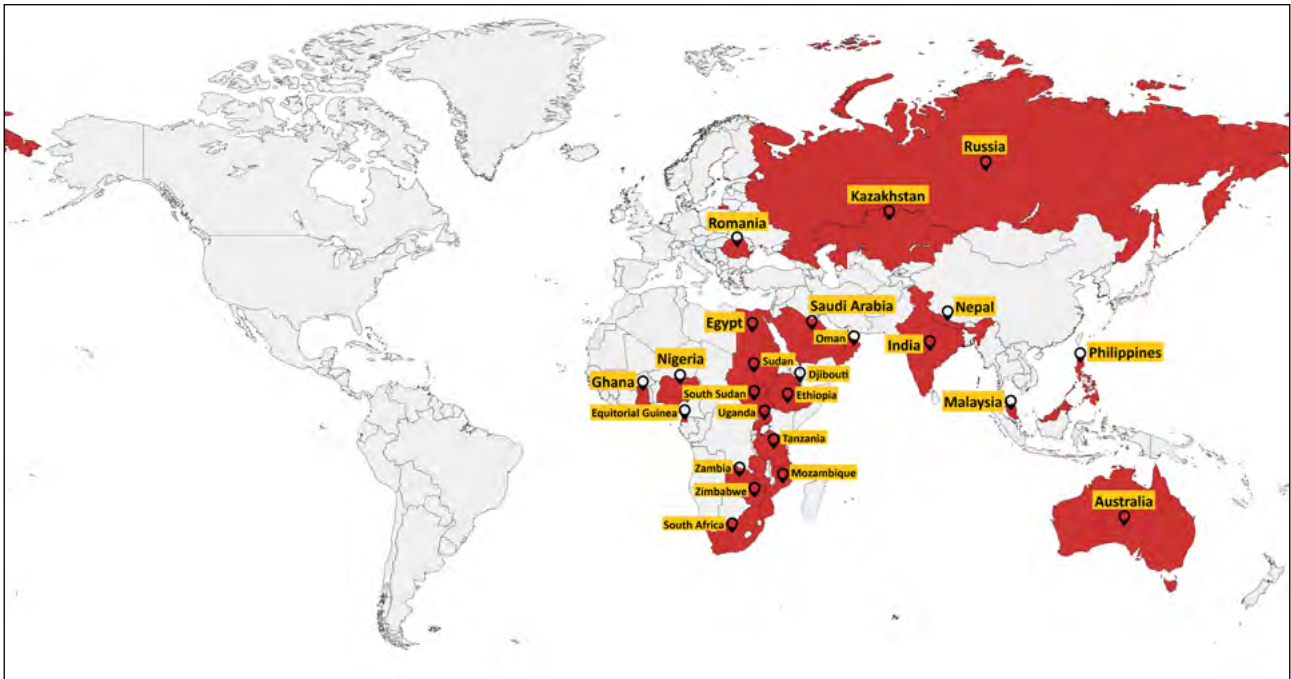
- PETN
- Ammonium Nitrate

SUSTAINABILITY

We are committed to **Green Mining**, developing eco-conscious solutions that minimize environmental impact without compromising power.

GLOBAL PRESENCE & EXPORT LEADERSHIP

SBL has the highest market share of Indian exports in emulsion cartridge, electronic detonators, and detonating cord.



TRUSTED BY

From local roots to global reach, our success is define by the enduring trust of our partners across every border.



THE SBL ADVANTAGE (CORE STRENGTH)

Backward Integration

Complete in-house
manufacturing
of 9 critical raw
materials for
sustainable quality
control and supply
reliability

Technical Collaboration

Mining technical
services team
of 150 and strategic
partnerships with
leading global
technology suppliers

R&D Focus

Dedicated research
team driving
innovations in
commercial
explosives, electronic
detonators and
insensitive munitions

Quality & Safety Certifications

Adherence to ISO,
PESO, SCOMET, and
Ministry of Defense
standards



CARTRIDGE EMULSION EXPLOSIVES

We provide an extensive range of globally consistent, high-quality packaged explosives tailored for the mining, quarrying, and construction sectors. Our comprehensive range ensures that operators have access to peak product performance across any application, regardless of geographic location. Engineered for excellent shock and heave energy, our solutions are designed to be safe, reliable, and user-friendly in the field.

ADVANTAGES

- Highly safe during manufacturing, transportation, storage, and handling, as the emulsion is classified as an oxidizer.
- Excellent resistance to water ingress, ensuring reliable performance in wet conditions.
- High velocity of detonation (VOD) and superior weight strength for efficient energy release.
- Low sensitivity to heat, contributing to enhanced operational safety.
- High viscosity and rigidity, providing improved charge stability and consistency.

SAFETY

- The emulsion is stable and does not detonate under standard impact or burning tests; however, it may detonate if initiated by high-energy materials such as detonators, dynamite, or aluminum powder.



CAP SENSITIVE EMULSION

LARGE DIAMETER CARTRIDGE EXPLOSIVES

(Booster / Primer Charge)

SHIPPING NAME EXPLOSIVES		CLASS / DIV.: 1.1D			UN NO: 0241			
Authorised Name of Product	PESO Brand ID	Diameter	Weight	No of cartridge in a box	Nominal density	Velocity of detonation	Relative weight strength	Relative Bulk strength
		(mm)	(grams)		(gm/cc)	(mtr/second)		
Neo Prime	620	65	1,250	20	1.15 ± 0.05	4,400 ± 300	118%	166%
		83	1,000	25				
		83	2,780	9				
		125	6,250	4				
		200	12,500	2				

REE is the effective energy relative to ANFO at a density of 0.85 g/cc. Energies based on ideal detonation calculation.

VOD of explosive depends on density, Hole diameter and degree of confinement. VOD quoted is unconfined condition.

APPLICATION

- Suitable for deep and long-hole blasting in opencast mines, quarrying, and hill cutting, delivering good fragmentation with excellent water resistance.
- Can also be used for plaster shooting and secondary blasting applications.

HOW TO USE

- Initiation shall be carried out using a No. 8 strength detonator along with detonating cord.
- The minimum permissible air gap under unconfined conditions is 2 cm.

PACKAGING

- Cartridges are packed in HDPE/LDPE film and further packed in 25 kg corrugated boxes.



CAP SENSITIVE EMULSION

SMALL DIAMETER CARTRIDGE EXPLOSIVES

SHIPPING NAME EXPLOSIVES		CLASS / DIV.: 1.1D			UN NO: 0241			
Authorised Name of Product	PESO Brand ID	Diameter	Weight	No of cartridge in a box	Nominal density	Velocity of detonation	Relative weight strength	Relative Bulk strength
		(mm)	(grams)		(gm/cc)	(mtr/second)		
Neo Gel 901	918	25	125	200	1.15 ± 0.05	4,500 ± 200	110%	165%
		32	200	125				
		40	400	62				
		50	625	40				

APPLICATION

- Ideal for opencast and underground metal mining, quarrying, well and shaft sinking, and tunneling applications, offering excellent water resistance.
- Suitable for plaster shooting and secondary blasting.

ADVANTAGES

- High-density explosive that performs reliably in wet and water-filled blast holes.

HOW TO USE

- The product is a high-strength, non-permitted emulsion explosive, intended for use as per approved initiation and blasting procedures.

PACKAGING

- Cartridges are packed in HDPE / LDPE film and further packed in 25 kg corrugated boxes.



SEISMIC EMULSION EXPLOSIVE

SHIPPING NAME EXPLOSIVES		CLASS / DIV.: 1.1D			UN NO: 0241		
Authorised Name of Product	PESO Brand ID	Diameter	Weight	No of cartridge in a box	Nominal density	Velocity of detonation	Hydrostatic Head
		(mm)	(grams)		(gm/cc)	(mtr/second)	
Neo Gel - 90 CPT	915	50	500	50	1.2 ± 0.05	5,000	58
		50	1,000	25			
		63	500	50			
		63	1,000	25			
		76	1,000	25			
		76	2,500	10			

APPLICATION

- Seismic explosive used for seismic exploration operations.
- Product density is engineered to ensure the cartridge sinks in water, enabling effective placement in water-filled shot holes.

ADVANTAGES

- High-strength explosive with high velocity of detonation (VOD) and excellent water resistance.
- Supplied in couplable tubes for ease of handling and reliable field deployment.

HOW TO USE

- Initiation is carried out using a No. 8 strength seismic detonator.
- Maximum sleeping time is approximately 8 weeks, subject to site conditions.

PACKAGING

- Cartridges are packed in Plastic Tubes and further packed in 25 kg corrugated boxes.



BULK EMULSION EXPLOSIVES

Bulk explosive systems / site-mixed systems are ideal for customers requiring large quantities of explosives. The materials are mixed on-site in specialized delivery trucks and are directly loaded into the shot holes by trained personnel. These systems are primarily intended for quarrying and open-pit mining operations, improving blast execution efficiency while enhancing safety, security, and environmental performance.

ADVANTAGES

- Excellent water resistance with an extremely safe and simple operating system.
- Highly stable product with long sleeping time (up to 10 days).
- Reduced inventory and minimal handling on the mine bench.
- Enables wider drilling patterns, improving overall cost efficiency.
- Completely fills blast holes without gaps or voids.
- Requires minimal manpower.
- Nominal site infrastructure required.
- Eliminates on-site disposal of packaging materials.

SAFETY

- Keep away from heat, sparks, open flames, and hot surfaces.
- Eliminate all potential sources of ignition; avoid activities that may cause impact, friction, sparks, or a sudden rise in temperature.
- Maintain product cleanliness and prevent contamination at all times.
- Not intended for storage; the product should be charged directly into blast holes at the point of use. If loaded holes are left sleeping for extended periods, adequate security arrangements must be ensured.

SPECIFICATIONS

PRODUCT NAME	NEO BULK
Category	Emulsion
Viscosity	More than 40,000 cps (Brookfield)
Sleeping time	More than 10 days
Sensitivity	Can be initiated by 0.2% of Cast Booster
Water resistance	Excellent
Air gap sensitivity	As per specifications
Application	Bulk Emulsion explosive sensitized by chemical gassing agents and stabilizers. NEO Bulk-901 is ideally suitable for Shovel as well as Dragline bench Blasting.

We are fully equipped to make on-site adjustments to the explosive mix, tailored to meet individual customer needs.



DETONATING FUSE / CORD

Detonating cord (or fuse) consists of a core of high explosive—typically PETN (Pentaerythritol Tetranitrate)—encased in layers of PP yarn, plastic sheathing (polyethylene jacket), and natural and synthetic fibres to ensure effective waterproofing and protection.

The cord is engineered with varying degrees of tensile strength, abrasion resistance, and flexibility, depending on the application. It serves as a safe, fast, and reliable means of initiating detonator-sensitive industrial explosives.

ADVANTAGES

- Relatively insensitive to detonation from heat, electrostatic discharge, or stray electrical energy.
- Excellent resistance to oil and water ingress, with superior abrasion resistance.
- High tensile strength for demanding field conditions.
- Flexible and easy to handle, including knotting.
- Ensures reliable non-electric initiation.
- Safer to use than conventional blasting caps, reducing handling and loading risks.
- Easy connection of branch lines, enabling efficient blast design.

SAFETY

- Protect from water in wet environments. While PETN absorbs moisture and becomes insensitive to detonation when wet, the detonating cord can still be initiated from a dry end.
- As an explosive product, detonating cord must be handled, stored, and transported with due care in accordance with applicable safety regulations.
- Sensitive to direct lightning strikes, severe impact or friction during handling, and extreme temperatures, particularly above 70°C.

SHIPPING NAME CORD DETONATING FLEXIBLE		CLASS / DIV.: 1.1D			UN NO: 0065		
Authorised Name of Product	PESO Brand ID	PETN / MT (grams)	DF per Spool (mtr)	No of Spool in a box (nos)	DF per box (mtr)	DIA (mm)	Tensile (kgs)
Neo Cord	1031	10	250	4	1,000	5.0 ± 0.1	55-65
		10	375	4	1,500		
Neo Cord 8	1153	08	375	4	1,500	4.7 ± 0.1	55-65
Neo Cord 12	1154	12	250	4	1,000	5.2 ± 0.1	60-65
		12	375	4	1,500		
Neo Cord 20	1155	20	125	4	500	6.2 ± 0.1	65-70

APPLICATION

- The Product is used for trunk line and down line for initiating explosive.
- Suitable for opencast mines, quarries, trenching and tunneling work.

HOW TO USE

- Initiation by Number 6 / Number 8 detonator.

PACKAGING

- Spools are packed in 28 kg corrugated boxes.

Velocity of Detonation ranges from 6500 to 7500 meter/second.



CAST BOOSTER

Cast boosters are designed to deliver high initiation and detonation pressure to a wide range of explosives and blasting agents. They are manufactured using high-density molecular explosives such as PETN and TNT, enclosed in a robust plastic shell.

Each booster features two longitudinal passages to accommodate a detonator or detonating cord, ensuring flexible initiation options. Cast boosters are used when a detonator alone does not provide sufficient energy to directly initiate the main explosive charge. In such cases, the detonator initiates the cast booster, which then transmits powerful shock waves to reliably initiate the explosive or blasting agent.

A recessed well at the base of the booster protects the detonating cord, signal tube, or lead wires from mechanical damage during handling and placement.

ADVANTAGES

- Cast Boosters have long storage life since it has no liquid ingredients and can be operated easily.
- Excellent shelf life, highly resistant to water and oil.
- Convenient to use and assemble.
- Do not contain Nitro-Glycerine, hence, less sensitive to shock and friction.
- High density and a high velocity of detonation maximise performance and makes them good primers and boosters.

SAFETY

- Should be transported, stored and handled with care.
- Avoid impact with solid surfaces or other Cast Boosters. This may lead damage that could lead to a misfire, or a premature initiation.
- Also, usage of boosters with detonators that cannot be completely fitted within the boosters should be strictly avoided as the detonator might be harmed during charging if not watched. This could lead to untimely detonation.
- Cast boosters may be used at temperatures up to 65°C.

SBL Energy offers Cast Boosters in 100gms, 150gms, 250 gms & 400 gms as per the needs of our CLIENTS.

SPECIFICATIONS

PRODUCT NAME	NEO BOOST
Classification	Class - 3, Div. – 2
Appearance	Yellowish Hard Solid Mass
Density	1.55 +0.05 gm/cc
VOD	7000 +/- 500 m/sec
Sensitivity	Cap Sensitive. Sensitive to No. 6 Detonator & Detonating Fuse
Water Resistance	Excellent
Weight per Cartridge	100gm/ 150gm/ 250gm/ 400gm
Package	25 Kgs. per box
Application	Initiation of non-cap sensitive charges in a bore hole, at any fixed point in a column of explosives charge. Used with ANFO, cartridges and bulk explosives to prime the boreholes.

PACKAGING

- Booster shells are packed in 24 / 25kg corrugated boxes.



DETONATORS

A detonator is an initiating device used to activate an explosive charge. Detonators can be initiated by three methods: electrical, mechanical, or chemical. They initiate explosives in a controlled manner by transmitting a signal from one point to another through electrical, mechanical, or chemical reactions. The timing of initiation is controlled either by electrical timing systems or by built-in chemical delay elements.

ORDINARY DETONATORS

SBL manufactures Ordinary Detonators under the brand name NEO OD (PESO ID: 1106). These are aluminum plain, non-electric ordinary detonators, commonly used in conjunction with safety fuse.

The detonators are of No. 8 strength, ensuring effective initiation of cap-sensitive explosives. Each detonator has a shell length of 37 mm and a shell diameter of 3.7 mm.

NEO OD detonators are manufactured in a state-of-the-art facility using advanced and highly controlled production techniques to ensure consistent quality and reliability. Initiation occurs when a safety fuse is properly inserted into the detonator.

APPLICATION

Detonators are ideally suited for use in dry blast holes and are widely used for boulder blasting, secondary blasting, and safe initiation in quarry operations, non-gassy underground and opencast mines, surface excavations, well sinking, road construction, and civil engineering works.

SAFETY

- Use a standard, approved-quality crimper to ensure proper and secure crimping of the detonator.
- Plain detonators contain sensitive components and must be handled with due care at all times.
- Use an adequate length of safety fuse to allow the shot firer sufficient time to reach a safe distance before detonation.
- Store in dry conditions at moderate temperatures, in accordance with applicable safety regulations.

PACKAGING

PRODUCTS	PESO BRAND ID	CATEGORY	PESO SIZE CODE	NO OF DETONATORS IN BUNCH / INNER BOX	NO OF BUNCH / INNER BOX	TOTAL DETONATORS IN A BOX
Neo OD	1106	Ordinary Detonator	DFQ	100	100	10,000



ELEMENTED CAPS

SBL manufactures Elemented caps under brand name 'Neo Al Elemented Det (Type A, B, C)'. These are aluminum elemented detonators commonly used in conjunction with shock tube and insulated electrical wires. Detonators are of strength of No. 8, ensuring effective initiation of cap sensitive explosives. Each detonator has shell length of 42mm to 95mm.

APPLICATION

- Elemented caps are ideally suited for use in crimping with shock tube / insulated electrical wires.

SAFETY

- Use a standard, approved-quality crimper to ensure proper and secure crimping of the detonator.

SPECIFICATIONS

PARTICULARS	UOM	Type A	Type B	Type C
Shell (Al) Length	mm	80 - 95	63.5 - 80	70 - 80
Inner Dia of Shell	mm	6.45	6.45	6.45
Outer Dia of Shell	mm	7.5	7.5	7.5
Delay Element	mm	6 - 29	6 - 29	6 - 29
Sealer Element	mm	13	13	13
IC Element	mm	8	8	8
Free Space	mm	26 - 35	26 - 35	26 - 35
Delay Time	ms	0 - 9000	0 - 200	200 - 600
ASA	gm	0.2 - 0.3	0.2 - 0.3	0.2 - 0.3
PETN	gm	0.8 - 1.2	0.6 - 0.8	0.6 - 0.8
ASA	mm	2 - 3	2 - 3	2 - 3
PETN	mm	20 - 23	20 - 23	20 - 23

NON-ELECTRIC DETONATOR

Non-electric detonators are widely used worldwide and are initiated by a shock wave transmitted through a shock tube from one detonator to another. They are used in both commercial blasting and military applications. The system consists of shock tubes connected to down-the-hole detonators (DTHD) and surface connectors, ensuring reliable initiation without electrical energy.

Compared to electric detonators, non-electric systems offer greater operational flexibility and enhanced safety, as they are insensitive to electricity, radio-frequency energy, and electromagnetic radiation—eliminating risks from stray currents and static electricity.

APPLICATION

NEO DET is used for large-scale blasting operations in mining and quarrying, particularly where fixed drilling and charge-loading patterns are followed. It is suitable for operations requiring both down-the-hole delay initiation and surface delay initiation within a single blasting circuit.

PRODUCTS

STLD (Surface Trunk line Detonator)

Neo Det is a non-electric surface trunk line detonator system designed to control millisecond delays between blast holes or rows through reliable surface sequencing. It consists of a durable shock tube of specified lengths attached to a colour-coded plastic connector housing a low-strength precision delay detonator. The connector block is engineered to accept multiple shock tubes, ensuring secure retention and flexible routing across the blast field. The system enables accurate hole-to-hole or row-to-row delay timing and is compatible with standard in-hole detonators, providing versatility in complex surface initiation designs.

DTHD (Down the Hole Detonator):

Neo det is a high-strength, non-electric millisecond delay detonator designed for reliable in-hole initiation across surface, underground, quarry, and construction blasting. It consists of an abrasion-resistant shock tube crimped to a precision delay detonator, with the free end sealed for moisture protection. The system offers multiple delay intervals for flexible timing design and is compatible with detonating-cord trunk lines and standard surface connectors. It is suitable for priming cap-sensitive explosives and boosters in a wide range of blasting environments.

In addition to the DTHD series, Neo-Det is also available in Short Period (SP) and Long Period (LP) delay configurations. These variants share the same construction, reliability, and compatibility as the DTHD detonators, offering extended flexibility in blast timing design across varied blasting conditions.

PRODUCTS

COMBIDET

NEO DET is a non-electric twin detonator system combining DTHD and STLD in one unit. A single shock tube coil is crimped with DTHD at one end and STLD at the other, enabling efficient down-the-hole and surface initiation. This integrated design reduces shock tube wastage compared to using DTHD and STLD separately.

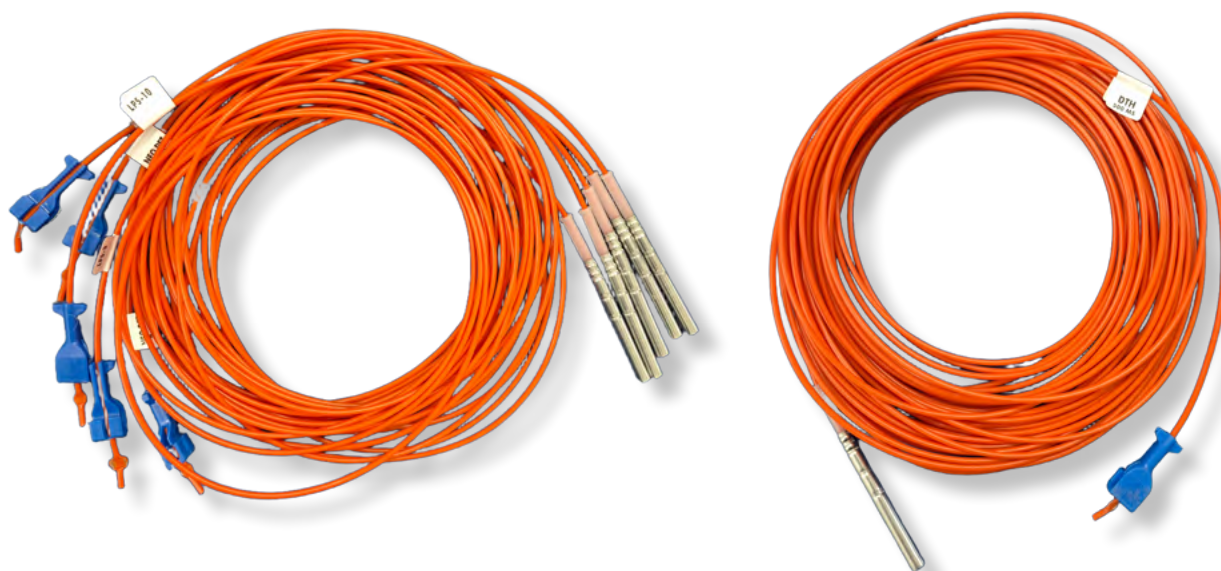
The STLD is housed in a connector capable of initiating up to six shock tubes. Each unit is clearly marked with length and delay identification stickers for easy handling. The detonator is of No. 8 strength, supports unlimited delay combinations, and is available in lengths of up to 50 meters, as per customer requirements. The delay timings of NEO DET are combinations of the following.

	NEO DTS (MS)	NEO STL (MS)
DELAY TIMING	125, 150, 175, 200, 225, 300, 350, 400, 450, 500	0, 17, 25, 42, 65, 85, 100

PESO Product ID - NEO DET – 632 / NEO DTS – 635 / NEO STL -636

PACKAGING

NEO DET units are initially packed in paper bags, which are then placed inside corrugated cardboard boxes. Each box contains 25 to 400 units, depending on the length of the shock tube.



ADVANTAGES

- Use of STLD for surface hook-up effectively eliminates air-blast noise.
- Provides adequate relief delay, reducing back break and creating a well-defined free face for subsequent drilling operations.
- Prevents disturbance of the stemming column and desensitization of the explosive column, resulting in higher explosive efficiency.
- Enables effective bottom initiation and improved control of ground vibration.
- Offers an extensive range of delay combinations, particularly beneficial for large-scale blasting operations.
- Eliminates the risk of misfires caused by leakage currents in conductive ore bodies or water-filled blast holes.



ELECTRONIC DETONATOR

NEO-EDET is a Fit-for-purpose electronic detonator system offering precise timing, security, and optimized blasting performance.

PRODUCT FEATURES

- Capacitor, logic & timing circuitry, and explosive housed in robust copper / Aluminum shell.
- Dedicated connector for reliable connection.
- High-strength design ensures dependable initiation of priming charges.

APPLICATION

- Quarrying
- Surface & underground mining
- Demolition
- Specialized blasting operations

ADVANTAGES

- Ultra-precise timing for improved fragmentation and enhanced vibration control.
- Complex blast patterns programmed and optimized through advanced computer software.
- Digital pre-testing of circuits and detonators with secure data logging and traceability.
- Enhanced safety and security, with initiation possible only through authorized firing equipment.
- Optimized blast efficiency, delivering compliant and consistent results.

SAFETY

- Handle only as per operational guidelines.
- Keep away from heat, sparks, and impact.
- Use only authorized firing equipment.
- Store under recommended conditions.

PACKAGING

NEO-EDET units are initially packed in polyethylene bags, which are then placed inside corrugated cardboard boxes. Each box contains 25 to 400 units, depending on the length of the wire.

NEO-EDET

Communication - Fully testable 2-way communication
Timing programmable, reprogrammable, pre-programmable
Traceability Unique ID number
Maximum time delay - 25000MS
Minimum increment - 1ms
Delay Accuracy-0.1% +/- 0.5MS
Operating temperature - from - 20°C to 80°C
Immunity to ESD-30kVKV & 2500PF
Anti-AC/DC-AC 220V no damage, DC48V no damage.
ESD resistance Compliant to EN 13763-13
Maximum number of modules 3000+ (Master-Slave Mode)



TNT (Trinitrotoluene)

SBL Energy Limited manufactures high-purity Trinitrotoluene (TNT), a globally recognized for its exceptional stability, safety, and reliable explosive performance across defense and commercial sectors. The product is supplied in a high-quality crystalline form, appearing as light-yellow flakes, free from mechanical impurities and moisture.

Naturally occurring bound flakes, if present, are easily separable by hand, ensuring the material is ready for immediate use in melt-pour production processes without affecting operational efficiency. With its low sensitivity to impact and consistent energy output, SBL Energy's TNT offers an optimal balance of safety and performance for demanding applications.

APPLICATION

- Used as a key ingredient in Pentolite cast boosters for the global mining and blasting industry.
- Serves as a base explosive for a wide range of military-grade explosive compositions and munitions.
- Suitable for melt-pour explosive manufacturing due to its thermal stability and uniform crystalline structure.
- Widely applied in defense, mining, and specialized energetic formulations requiring dependable performance and controlled sensitivity.

SPECIFICATIONS

Appearance	Light Yellow Flakes
Colour	Not Darker than No.30257 supersedes by AMS-STD-595
Odor and Taste	Odourless, Tasteless
Molecular Weight	227.13 g/mol
Water Solubility	0.013 g/l at 20°C
Moisture %	0.10 Max
Acidity % (as H ₂ SO ₄)	0.10 Max
Alkalinity	None
Solidification Temperature	≥80.2°C
Boiling Point	240°C
Flash Point	290°C to 300°C
Vapor Pressure	1.98 X 10 ⁻⁴ mm Hg
Density	1.654 g/cm ³ at 20°C
Detonation Velocity	7000 m/s at 1.62 g/cm ³
Insoluble in (Acetone / Benzene / Toluene)	0.10 Max

STORAGE

TNT shall be stored in well closed, fireproof Magazines and as per local statutory rules and regulations. If stored Properly, TNT will have very longer shelf Life.

PACKAGING

TNT is packed and supplied in 25kg corrugated boxes with 2 plastic liners inside.



HMX (High Melting Explosive) / OCTOGEN

SBL Energy Limited is a trusted manufacturer of premium-grade HMX, engineered to meet the stringent performance requirements of strategic defense and advanced energetic systems. Our state-of-the-art manufacturing processes ensure consistent purity, high detonation velocity, and exceptional energy output, making our HMX the preferred choice for applications where precision, reliability, and power are mission-critical.

Designed for integration into sophisticated explosive formulations, SBL Energy's HMX provides a robust and dependable foundation for high-performance shaped charges and military-grade compositions, supporting tactical superiority in the most demanding operational environments.

APPLICATION

- Melt-cast and pressed military charges.
- Cast-cured charges for Insensitive Munitions (IM).
- Pyrotechnic systems (cap-relay, cutting cords, detonators).
- Initiation and main charges for oil-well perforating systems.
- Formulation of plastic bounded explosive and melt castable composition.

ADVANTAGES

- Very high detonation velocity.
- Excellent thermal stability.
- High density and energy content.
- Consistent and predictable performance.
- Suitable for advanced and specialized applications.

HANDLING AND OPERATIONAL SAFETY

- Use only approved personal protective equipment (PPE) specifically designed for handling high-energy materials.
- Handle with care to avoid physical damage to packaging; ensure containers remain sealed and secure at all times.
- Prevent contact with incompatible materials and operate within a strictly controlled handling environment.

SPECIFICATIONS

Appearance	White Crystalline Powder
Odor	Odourless
Purity %	Grade I - >93% and Grade II - >98%
Chemical Formula	$C_4H_8N_8O_8$
Molecular Weight	296.2 g/mol
Minimum % Water	15%
Boiling point / Boiling range	276°C - 280°C
Deflagration point	287°C
Detonating Velocity	9100 m/s
Danger of Explosion	Danger of explosion due to impact, friction, fire and other sources of ignition. The heat can cause an explosion.

ADVANCED STORAGE REQUIREMENTS

- Store in a dry, well-ventilated area at stable ambient temperatures to preserve chemical integrity.
- Keep strictly isolated from heat sources, open flames, sparks, and other potential ignition points.
- Avoid exposure to direct sunlight; store away from strong acids, alkalies, and reactive chemicals.

PACKAGING

HMX is packed and supplied in 8kg corrugated boxes with 2 anti-static liners inside for enhanced protection and safe transportation.

RDX (Cyclotrimethylene Tri nitramine)

SBL Energy Limited manufactures RDX (Cyclotrimethylene Tri nitramine) in accordance with stringent quality and process controls to meet defense and strategic application requirements. The product is characterized by high detonation velocity, controlled particle size distribution, uniform morphology, and consistent energetic performance.

RDX supplied by SBL Energy Limited is suitable for use in high-performance explosive formulations, where reliability, repeatability, and compliance with safety and storage norms are critical.

APPLICATION

- High-performance military munitions.
- Plastic bonded explosives (PBX).
- Cast-cured and pressed explosive formulations.
- Propellants and rocket motor compositions.
- Boosters and main charges in defense systems.

ADVANTAGES

- High explosive energy output.
- Consistent detonation velocity.
- Uniform particle morphology.
- Excellent compatibility with standard binders and additives.
- Suitable for both pressed and cast-cured formulations.

SPECIFICATIONS

Appearance	White or Brown Crystalline Powder
Odor	Odourless
Concentration	> 95 %
Chemical Formula	$C_3H_6N_6O_6$
Molecular Weight	222.12 g/mol
Solubility in water at 20°C	38.4 – 38.9 mg/L; 60 mg/L
Melting point / Melting range	200°C
Flash Point	229°C
Decomposition Temperature	234°C
Density	0.85 – 1.00 g/cm ³
Detonating Velocity	8750 m/s
Danger of Explosion	Danger of explosion due to impact, friction, fire and other sources of ignition. The heat can cause an explosion.

HANDLING AND SAFETY

- Personnel shall use approved personal protective equipment (PPE) during handling and processing.
- Adequate precautions shall be taken to prevent electrostatic discharge.
- The product shall be kept away from open flames, hot surfaces, sparks, and all ignition sources.
- Containers shall be handled carefully to avoid physical damage and contamination.
- Contact with incompatible materials shall be strictly avoided.

ADVANCED STORAGE REQUIREMENTS

- Store in original, sealed containers only.
- Storage shall be in a dry, well-ventilated magazine at stable ambient temperatures.
- Where specified, the product shall be maintained in wetted condition to ensure stability and fire resistance.
- Protect from direct sunlight and environmental exposure.
- Storage facilities shall be isolated from drains, sewers, and incompatible chemicals, particularly strong acids and alkalies.

PACKAGING

RDX is packed and supplied in 8kg corrugated boxes with 2 anti-static liners inside.

COMPLIANCE

Manufactured in compliance with applicable explosive safety regulations and quality assurance systems. Documentation and certification can be provided as per defence tender, DGQA, or export control requirements.

PYRO DEVICES

A flare is an IR (Infra-Red) countermeasure that is commonly used to protect target platforms against IR heat-seeking SAM (Surface-to-Air Missile) or AAM (Air-to-Air Missile) threats. It is not only employed to protect aircrafts, such as airplanes or helicopters, but also naval platforms from anti-ship threats. A typical flare is composed of a cartridge case, which protects the flare before the ejection and acts as launching tube, an ejector charge, which allows the ejection of the flare from the dispenser, a bore safety and initiator, which ignites the pyrotechnic pellet after the ejection and a closure cap, which seals the case. The pellet has in charge the generation of the IR radiation by means of pyrotechnic or pyrophoric charges. Burning temperature and emissivity influence the radiated energy and spectral distribution. MTV (Magnesium/Teflon/Viton) flares are the simplest and most common type, burning at very high temperature and generating a strong radiant intensity.

ADVANTAGES

High burning temperature allows strong radiant intensity and high energy output. The flare can act as an effective decoy by producing an infrared signature. Different compositions such as MTV provide intense radiation, making them suitable against IR-guided threats.

APPLICATIONS

Used as defensive countermeasures on aircraft including airplanes and helicopters, and on naval platforms, to protect against heat-seeking missile threats such as SAM, AAM, and anti-ship missiles.

AMMUNITION

Warheads consist of High explosive charges like RDX/HMX inside a container which can be a pre-fragmented metal container. To give high energy output, the explosive is pressed into shaped charges. Different polymers like polypropylene, polyethylene, polyurethane, polystyrene, wax, HTPB, etc are used as binders. HMX/RDX crystals are encapsulated with these polymers and stored as compounded products of HMX/RDX. The pellet is bonded to the hardware with prescribed bonding resin, and the warhead is then marked and packed.

ADVANTAGES

Use of high explosive materials such as RDX and HMX provides high energy output. Pressed shaped charges and polymer-bonded compositions allow the explosive to be formed and handled as compounded products.

APPLICATIONS

Used as ammunition warhead sections where high explosive output and pre-fragmented containment are required.

PETN (Penta Erythritol Tetra Nitrate)

PETN (Pentaerythritol Tetranitrate) is a secondary explosive that appears as white, hexagonal crystals. It is characterized by high detonation velocity and reliable performance, making it suitable for use in initiating and transfer applications rather than as a standalone high explosive.

APPLICATION

- Used as the explosive core of detonating cord.
- Employed as a component in composite explosives.
- Used as the base charge in blasting caps.
- Supplied as cast primers, including shaped charges, for specialized blasting and demolition applications.

ADVANTAGES

- Can be stored for longer periods without significant deterioration compared to nitro-glycerine or nitrocellulose.
- Easily initiated using a detonator or detonating cord.
- Soft, paste-like consistency allows easy shaping for specialized applications.

SPECIFICATION

Bulk Density (Tapped) g/ml	1.00 ± 0.1
Moisture % by mass, Max (Dry)	0.1
Nitrogen Content (min %)	17.5%
Melting Point	139°C to 142°C
Stability (Able Heat Test) at 76.5 ± 1.5°C, minutes, min	60
Total Acidity	0.01% Max
Insoluble in Acetone	0.05% Max
Ash Content	0.005% Max
Sleeve Analysis (Pass Through)	
14 Mesh BSS Mesh % Max	100%
36 Mesh BSS Mesh % Max	>55%
10 Mesh BSS Mesh % Max	<10%
Flow Rate g/min	160

SAFETY

- Dropping or ignition will typically not cause an explosion, as PETN is less sensitive than primary explosives; however, it is more sensitive to shock and friction compared to many other secondary explosives.
- Under certain conditions, a deflagration-to-detonation transition (DDT) may occur.
- PETN is considered less stable than several other commonly used military explosives and must therefore be handled with appropriate precautions.

PACKAGING

PETN (25% moisture) is securely packed in double polythene bags, further enclosed in an HDPE woven bag and finally packed in a corrugated cardboard case.

The packaging is approved by PESO (India) and the Indian Institute of Packaging (IIP), Mumbai for UN-certified transport of hazardous goods.

- Net weight per box: 10 kg
- Gross weight per box: 15 kg
- Container loading capacity: Up to 15 MT (approx. 1,500 boxes per container)



AMMONIUM NITRATE (HDAN & LDAN)

SBL Energy Limited through its subsidiary supplies High Purity Ammonium Nitrate sourced from approved and reputed manufacturers, meeting stringent quality and consistency requirements for explosive and industrial applications. The product is offered in both High Density Ammonium Nitrate (HDAN) and Low Density Ammonium Nitrate (LDAN) grades to suit varied blasting system requirements.

The material is characterized by uniform granulation / prill structure, controlled density, low impurity levels, and consistent performance, ensuring reliable downstream processing and predictable blast outcomes.

APPLICATION

- Bulk and packaged explosive formulations.
- ANFO and site mixed blasting systems.
- Quarrying and opencast mining operations.
- Civil infrastructure and controlled blasting applications.

PRODUCT GRADES

HDAN

(High Density Ammonium Nitrate)

- Dense, mechanically strong granules.
- Suitable for bulk explosive systems and heavy column charges.
- Provides structural integrity during transportation, handling, and loading.

LDAN

(Low Density Ammonium Nitrate)

- Porous prills with high fuel oil absorption capacity.
- Optimized for ANFO and site mixed explosive formulations.
- Ensures efficient energy release and uniform detonation performance.

PACKAGING

Supplied in 1,000 kg and 1,250 kg FIBC bulk bags, suitable for mechanized handling and bulk operations.

STORAGE

- Store in a dry, well ventilated environment.
- Protect from humidity and moisture ingress to maintain free flowing properties and consistent performance.
- Ensure bags are kept sealed and protected from physical damage during storage.

SPECIFICATIONS

High Density Ammonium Nitrate	
Colour	White
Organic Coating	No Coating / May contain some traces
Content of NH_4NO_3 in dry matter	Min 98.0
N% (as NH_4NO_3) %	Min 34.0
Magnesium Nitrate Content as $\text{Mg}(\text{NO}_3)_2$ %	Min 1.0
Water content by Carl's Fisher Titration %	Max 0.6
Carbon content in carbon equivalent %	Max 0.2
pH factor of 10% solution %	Min 5.0
Water insoluble %	Max 0.005
Free flowing, free from any foreign impurities substances	Danger of explosion due to impact, friction, fire and other sources of ignition. The heat can cause an explosion.
(Friability)	100%

Low Density Ammonium Nitrate	
Colour	White
Organic Coating	No Coating / May contain some traces
Total Weight share nitric and ammonium recalculated for NH_4NO_3 in dry substance	Min 99.5%
Total weight shares nitric and ammonium recalculated for Nitrogen in dry substance	Min 34.4%
Water content by Carls Fishers titration	Max 0.2%
pH factor of 10% solution %	Min 4.5
Oil absorption	Min 11%
Bulk density g/cm ³	0.7 – 0.8
Granulometry	
< 1 mm	Max 5%
> 2 mm	Max 15%



BLASTING CONSULTING & TECHNICAL SERVICES

Maximise Efficiency | Optimise Costs | Ensure Compliance

SBL Energy Limited delivers integrated blasting consultancy and technical services designed to maximise productivity, optimized cost efficiency and ensuring safety. Supported by a specialised team of mining engineers, we leverage modern analytical tools and field expertise to ensure that each blast is executed with precision and consistency.

From explosive selection and blast design to on site execution and performance evaluation, SBL provides tailored solutions aligned with site specific geology, production targets, and regulatory requirements.

We Assist customers in



TECHNICAL EXPERTISE & FIELD SUPPORT

Our Sales and Technical Engineers provide customers with comprehensive guidance on blast design, measurement techniques, and field application, supported by structured safety training and correct product handling practices. Services are strengthened through rock face analysis, drilling evaluation, and site specific technical assessments, enabling reliable and repeatable blasting outcomes.

BLAST DESIGN SERVICES

SBL provides site specific blast designs based on geological conditions, production requirements, and vibration constraints. Advanced fragmentation modelling and vibration control techniques are applied to reduce secondary blasting and improve downstream process efficiency.

TECHNICAL BLAST MANAGEMENT SERVICES

At SBL, we focus on converting complex blasting challenges into controlled, cost effective operations. Our Technical Services Wing, comprising 150+ qualified mining engineers, bridges engineering theory with real world execution to enhance operational efficiency and blast performance.

OUR CORE SPECIALIZATIONS

- **Blast & Cost Optimization:** Data driven designs to reduce drilling and blasting costs.
- **Controlled Blasting:** Precision techniques to protect infrastructure and maintain site integrity.
- **Fragmentation Analysis:** Engineered rock sizing for efficient loading, hauling, and crushing.
- **Muck-pile Profiling:** Optimised heave and throw for improved diggability and equipment productivity.
- **Vibration Control:** Monitoring and mitigation to ensure environmental and statutory compliance.

HIGH-IMPACT TECHNICAL SOLUTIONS

- **Dragline Bench & Large Scale Blast Optimisation:** Optimisation of power factors for high volume excavation.
- **Knowledge Transfer & Training:** Custom training programs for Blasting Officers and Managers covering safety, design methodology, and best practices.
- **Continuous Technical Development:** Adoption of global advancements in blasting chemistry, digital tools, and execution methodologies.

EXPLOSIVE REGULATORY COMPLIANCE

Our team ensures full compliance with explosives safety regulations and statutory requirements, including permitting support, site inspections, documentation, and audit assistance. This enables uninterrupted operations while minimising legal and operational risk.

WHY CHOOSE SBL?

- **Expert Team:** 150+ specialised mining engineers.
- **Integrated Approach:** Seamless alignment of products and technical services.
- **Proven Results:** Measurable reductions in blasting costs and improved performance.
- **Compliance Focus:** Safety driven, regulation compliant operations.

SBL Technical Services are available on request, providing customers with a scalable engineering partner dedicated to long term operational success.



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